Deployment Optimization of Dual-Functional UAVs for Integrated Localization and Communication

Zheyuan Yang, *Graduate Student Member, IEEE*, Suzhi Bi, *Senior Member, IEEE*, and Ying-Jun Angela Zhang, *Fellow, IEEE*

***Abstract*— In emergency scenarios, unmanned aerial vehicles (UAVs) can be deployed to assist localization and communication services for ground terminals. In this paper, we propose a new integrated air-ground networking paradigm that uses dual-functional UAVs to assist the ground networks for improving both communication and localization performance. We investigate the optimization problem of deploying the minimal number of UAVs to satisfy the communication and localization requirements of ground users. The problem has several technical difficulties including the cardinality minimization, the non-convexity of localization performance metric regarding UAV location, and the association between user and communication terminal. To tackle the difficulties, we adopt D-optimality as the localization performance metric, and derive the geometric characteristics of the feasible UAV hovering regions in 2D and 3D based on accurate approximation values. We solve the simplified 2D projection deployment problem by transforming the problem into a minimum hitting set problem, and propose a low-complexity algorithm to solve it. Through numerical simulations, we compare our proposed algorithm with benchmark methods. The number of UAVs required by the proposed algorithm is close to the optimal solution, while other benchmark methods require much more UAVs to accomplish the same task.**

***Index Terms*— Wireless localization, integrated localization and communication (ILAC), unmanned aerial vehicle (UAV), geometric deployment.**

Manuscript received 2 September 2022; revised 30 January 2023; accepted 25 April 2023. Date of publication 10 May 2023; date of current version 12 December 2023. This work was supported in part by the National Natural Science Foundation of China under Project 62271325; in part by the Major Key Project of Peng Cheng Laboratory (PCL) Department of Broadband Communication; in part by the Guangdong Basic and Applied Basic Research Foundation under Project 2022A1515010973 and Project 2022A1515011219; in part by the Key Project of Department of Education of Guangdong Province under Grant 2020ZDZX3050; in part by the Shenzhen Science and Technology Program under Project 20220810142637001 and Project JCYJ20210324093011030; in part by the Open Research Project Program of the State Key Laboratory of Internet of Things for Smart City (University of Macau) under Grant SKL-IoTSC(UM)-2021-2023/ORPF/A03/2022; in part by the General Research Fund under Project 14201920, Project 14202421, and Project 14214122; and in part by the Area of Excellence Scheme from the Research Grants Council of Hong Kong under Project AoE/E-601/22-R. The associate editor coordinating the review of this article and approving it for publication was G. Fodor. *(Corresponding author: Suzhi Bi.)*

Zheyuan Yang and Ying-Jun Angela Zhang are with the Department of Information Engineering, The Chinese University of Hong Kong, Hong Kong (e-mail: yz019@ie.cuhk.edu.hk; yjzhang@ie.cuhk.edu.hk).

Suzhi Bi is with the College of Electronics and Information Engineering, Shenzhen University, Shenzhen 518060, China, and also with the Department of Broadband Communication, Peng Cheng Laboratory, Shenzhen 518066, China (e-mail: bsz@szu.edu.cn).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TWC.2023.3273159.

Digital Object Identifier 10.1109/TWC.2023.3273159

I. Introduction

A. Motivations

MODERN 5G communication networks with high speed and low latency are expected to cover most urban areas. New applications enabled by 5G such as remote healthcare, Internet of Vehicles, smart homes, industrial control, and environmental monitoring will bring unprecedented intelligent service experience to citizens [1], [2], [3], [4]. However, urban 5G networks are under the risk of regional service interruption in case of earthquakes, floods or other emergency accidents [5]. Besides, in remote areas such as mountainous environments and wild fields, it is difficult to guarantee smooth communication when performing terrain survey, reconnaissance and rescue operations due to the high deployment cost and insufficient coverage of communications infrastructure. Recently, unmanned aerial vehicle (UAV) technology has emerged as an important solution to assist communication in emergency, where UAV-mounted aerial base stations (BSs) can be flexibly deployed to cover ground users with high-speed and reliable line-of-sight (LoS) communication services [6].

Besides reliable communication services, ground terminals such as rescue devices also demand accurate positioning services in emergency scenarios. The global navigation satellite system (GNSS) can provide satisfactory positioning services in open-sky environments [7]. However, under severe obstructions and scattering in mountainous or disaster environments, the positioning accuracy of GNSS plummets dramatically (e.g., from 5-10 meters root mean square error to tens of meters) and the positioning service suffers from frequent interruptions [8]. Besides, the GNSS has relatively low positioning accuracy in vertical dimension, and thus it cannot provide accurate three-dimensional (3D) localization services required in many emergency scenarios [9]. In addition to GNSS, existing terrestrial cellular networks have independent localization capabilities, where terrestrial BSs can serve as anchor nodes (ANs) to support a variety of localization methods such as angle-based localization [10] and observed time difference of arrival (OTDoA) positioning [11]. Achieving high 3D localization accuracy requires not only sufficient number of ANs, but also spatial diversity in the deployment of ANs [12]. For instance, to attain high vertical positioning accuracy, the anchors should differ noticeably in amplitude. When the ground ANs are co-planar, UAVs can be quickly and flexibly deployed as temporary aerial ANs to improve

1536-1276 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.

the overall localization accuracy, especially in the vertical dimension. Existing UAV deployment solutions mostly use UAVs for either communication or positioning purpose, and lack efficient coordination with terrestrial networks. As a result, a large number of UAVs are needed to ensure both communication and positioning performance in a large area. In practice, this could lead to large networking delay, high control complexity and signalling overhead.

To minimize the UAV deployment cost of emergency networks, we propose a dual-functional UAV paradigm that utilizes UAVs as both air-based ANs and BSs to provide integrated localization and communication (ILAC) services to ground UEs. Specifically, we consider to deploy multiple dual-functional UAVs working collaboratively with terrestrial BSs to extend the communication and localization service coverage. The key design problem lies in the potential conflicting effects of UAV positions on the communication and localization performance of ground UEs. Intuitively, as an aerial BS, a UAV needs to be deployed close to the ground UEs for stronger communication link quality; as an aerial AN, however, such a close-to-ground UAV deployment may cause large vertical localization error due to its similar altitude with terrestrial anchors. To meet the diverse service requirements of ground UEs, the deployment of UAVs should coordinate with terrestrial BSs to achieve a balanced communication and localization performance.

To this end, this paper aims to answer the following two key questions:

Q1: For a UE at a given target location, how to determine the position of a single dual-functional UAV *in an analytical form*, such that the UAV and ground BSs can collaboratively satisfy the communication and localization performance requirements of the UE.

Q2: For a set of target UEs distributed over a large area, how to deploy minimum number of dual-functional UAVs *within a short computational time*, such that the integrated air-ground network can meet the communication and localization performance requirements of all the UEs.

To understand the technical challenges of the above questions, we review in the following some related work using UAVs to provide communication and localization services.

B. Related Work

1) UAV-Assisted Communication: There have been extensive studies in recent years on employing UAVs to assist terrestrial communications. For instance, [13] outlines the characteristics and potential advantages of an integrated air-ground heterogeneous network architecture. UAVs can be deployed in mobile hot-spots to offload cellular traffic, as aerial relays to tackle obstructions in the ground, as transceivers to collect/disseminate data from/to massive IoT devices in rural areas, and as temporary aerial stations in emergency, etc. UAV deployment is one key design problem in UAV-assisted communications. Depending on the mobility of UAVs, existing studies are mainly divided into two streams: one optimizes the fixed locations of rotary-wing UAVs and the other designs the trajectories of mobile UAVs, including both rotary-wing and fixed-wing UAVs. In this paper, we focus on the former topic to deploy dual-functional UAVs at fixed locations.

Considering a simplified one-dimensional (1D) scenario, the authors in [14] derive the formula of LoS probability between the UAV and the ground users, and optimize the UAV altitude to maximize the UAV communication coverage. For multiple UAVs, the authors in [15] study the optimal altitude of UAVs as relays to minimize the end-to-end outage probability and bit error rate (BER). References [16] and [17] further consider the UAV placement for communication in 2D scenario when the UAV altitude is fixed. Reference [16] aims to minimize the number of UAVs to cover a group of ground users by sequentially deploying the UAVs along a spiral path from area perimeter toward the center. Reference [17] jointly optimizes the UAV 2D locations and user transmission power to maximize the user communication rate to both UAVs and multiple terrestrial BSs. Furthermore, recent studies [18], [19], [20] investigate how to deploy UAVs in 3D space to improve the communication performance. Considering using a single UAV as a relay, [18] minimizes the decoding error probability in latency-critical scenarios by optimizing the location and power of the UAV. Using multiple UAVs as relays, [19] searches for the optimal UAV locations to maximize the system communication rate. Reference [20] investigates the problem of determining 3D placement and orientation of the minimal number of UAVs to guarantee the LoS coverage and the signal-to-noise ratio (SNR) between the UAV-UE pairs.

2) UAV-Assisted Localization: Besides GNSS, a UE can also estimate its location from the positioning reference signals sent by cellular networks, such as received signal strength (RSS), time of arrival (ToA) and time difference of arrival (TDoA) [21]. Among them, location estimation using TDoA measurements is one popular method used in 4G/5G standard for high-precision localization. The estimation is based on the principle of linear least-square estimation, where at least four ANs are required to obtain an unbiased 3D location estimation. Additional ANs can provide redundancy in the estimation to counter noisy measurements.

Given the deployment of ANs, there have been extensive studies to optimize the resource allocation such as power and bandwidth allocation to minimize the localization error. References [22] and [23] study the resource allocation for UAV-assisted vehicle localization using ToA method. Both studies obtain the global optimal solution based on semi-definite programming (SDP) that minimizes the Cramer-Rao lower bound (CRLB) of the localization error. However, minimizing the localization error with respect to UAV anchor deployment is difficult due to the non-convexity of CRLB regarding the UAV position. The current research on UAV deployment and trajectory optimization mainly adopts exploration algorithms, such as genetic algorithms, predetermined-pattern trajectory optimization and particle swarm optimization [24], [25], [26]. Reference [24] plans a static path for a single UAV by selecting a subset of way-points from a predetermined point set to maximize the localization precision. In [25], the authors jointly optimize the UAV positions, power and bandwidth allocation to improve the localization accuracy of vehicles by Taylor expansion-based

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.

approximate searching algorithm. Based on RSS measurements, [26] designs the UAV trajectory to assist 3D map estimation using particle swarm optimization. In general, the existing exploration-based methods lack theoretical analysis on the optimal UAV deployment. As a result, the obtained solution often cannot guarantee localization performance and may induce high computational complexity in emergent UAV deployment applications.

3) Summary: To summarize, communication-oriented UAV placement is well-understood from 1D to 3D scenarios. However, localization-oriented UAV placement lacks theoretical characterization and efficient optimization tools. In addition, most work studies the communication and localization problems separately, which may lead to high deployment cost and delay in emergency network. Reusing one UAV for providing both communication and localization services is an effective solution to reduce the network deployment cost, which however is also more challenging considering the potential performance conflict in the dual-functional UAV deployment. To achieve guaranteed quality of service (QoS), we need both theoretical analysis and low-complexity algorithm design for the deployment of dual-functional UAVs.

C. Contributions

In this paper, we study the optimal deployment problem of dual-functional UAVs that operate collaboratively with ground BSs to provide both communication and localization services to ground UEs. In particular, we are interested in deploying minimal UAVs to satisfy the communication and localization performance requirements of UEs at a set of target locations. Our contributions are summarized as follows.

1) Low-Cost Networking With Dual-Functional UAVs: We propose a new integrated air-ground networking paradigm that uses dual-functional UAVs to assist the ground networks for improving both communication and localization services. We consider a UE using OTDoA method to estimate its own location from the positioning signals sent by a UAV and three ground BSs. Besides, each UE communicates to either a UAV or a BS that yields the highest data rate. The proposed dual-functional UAV scheme significantly reduces the cost and delay in the network deployment under emergency.

2) Minimal UAV Deployment Problem Formulation: We formulate a minimal UAV deployment problem that optimizes both the number and locations of UAVs to meet the communication and localization requirements of ground UEs. The problem is very challenging because of the intractable cardinality minimization, the combinatorial UAV-UE communication association, and non-convex UAV location optimization. We solve the problem in two steps. First, we derive the closed-form expression of feasible UAV location to meet the dual performance requirements of each UE. Then, we transform the minimum deployment problem into an equivalent graphical form and design an efficient algorithm accordingly.

3) Geometric Characterization of Feasible UAV Location: We first analyze the feasible deployment location of a UAV that meets the localization accuracy requirement of a UE. Instead of conventional CRLB performance metric, we propose to use *D-optimality* as the key localization accuracy metric for analytical tractability. We show that, under the *D-optimality* metric, the feasible location of the UAV can be characterized as a second-order cone in 3D space. Meanwhile, given a fixed altitude, it reduces to an ellipse in 2D projection plane. We derive the closed-form expressions of both the 3D and 2D feasible regions of the UAV.

4) Efficient Graphical Solution Algorithm: With the geometrical characterization of feasible region, we show that the minimal deployment problem can be equivalently transformed to a *minimum hitting set* problem, which is a classical NP-complete problem that lacks efficient solution. Based on the equivalent graphical formulation, we propose a low-complexity approximate algorithm to tackle the NP-hardness of the problem in large-size emergent networks.

Simulation results show that the proposed method achieves close-to-optimal performance with much lower computational complexity. Besides, compared with conventional scheme with four ground anchors to locate a 3D target, we find that the proposed integrated air-ground network paradigm significantly improves the localization accuracy in both horizontal and vertical directions. This is because the deployed UAV improves not only the altitude diversity, but also the anchor-target localizing link strength especially when the NLoS noise is large in ground-to-ground channels. Overall, the proposed UAV deployment method provides an efficient solution to guarantee both communication and localization performance requirements.

The rest of paper is organized as follows. Section II introduces the system model of UAV-assisted air-ground ILAC scheme and we formulation the minimum UAV deployment problem in Section III. In Section IV and Section V, we analyze the feasible UAV hovering regions and propose a low-complexity deployment algorithm. Section VI presents the simulation results to evaluate the algorithm performance. Lastly, Section VII concludes the paper.

II. System Model

As shown in Fig. 1, we consider an emergency network consisting of 3 terrestrial BSs, U rotary-wing UAVs and K UEs. The locations of the n-th BS, the u-th UAV and the k-th UE are denoted as $\boldsymbol{b}_n = [x_n^b, y_n^b, h_n^b,], n \in \mathcal{N} \triangleq \{1,2,3\}$, $\boldsymbol{b}_u = [x_u^a, y_u^a, h_u^a], \forall u \in \mathcal{U} \triangleq \{1,2,\ldots,U\}$ and $\boldsymbol{m}_k = [x_k, y_k, h_k], \forall k \in \mathcal{K} \triangleq \{1,2,\ldots,K\}$, respectively.[1] The UAVs and BSs collaboratively provide communication and localization services to ground UEs.

A. Communication Model

Considering the occasional blockage between the UAVs and the ground UEs, we adopt the commonly used probabilistic LoS channel model to determine the large-scale attenuation for the ground-to-air (G2A) links [14]. The probability of geometric LoS between the UAV and UE depends on statistical

[1]Notice that the accurate user location is not known and to be estimated. The location $\boldsymbol{m}_k$ is in fact a target user location to receive communication and localization service, e.g., from initial coarse location estimation. A naive way to set the target user locations is dividing the area of interest into equally separated points. Without causing confusions, we use target user location and user location interchangeably in this paper.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.

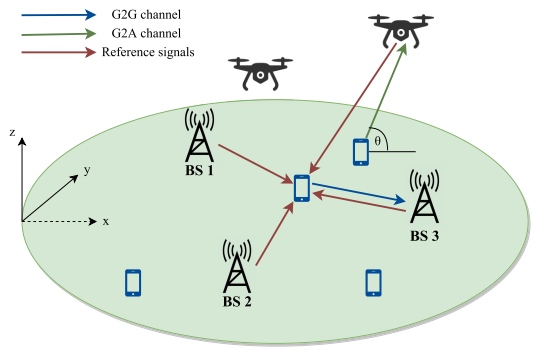


Fig. 1. System model of the air-ground ILAC network.

parameters related to the environment and the elevation angle. Specifically, we denote the LoS probability between the UAV at location $\boldsymbol{b}_u$ and UE k as $\mathbb{P}(LoS, \theta_{ku})$, which can be approximated as a modified sigmoid function of the following form [14]

$$\mathbb{P}(LoS, \theta_{ku}) = \frac{1}{1 + e_1 \exp\left(-e_2(\theta_{ku} - e_1)\right)}, \tag{1}$$

where e_1 and e_2 are environment-related parameters. θ_{ku} is the elevation angle given by

$$\theta_{ku} = \frac{180}{\pi} \arcsin\left(\frac{|h_u^a - h_k|}{||\boldsymbol{b}_u - \boldsymbol{m}_k||}\right), \tag{2}$$

and $\theta_{ku} \in [0°, 90°]$. Accordingly, the expected channel power gain is equal to

$$\begin{aligned} g_{ku} &= \frac{\mathbb{P}(LoS, \theta_{ku}) + (1 - \mathbb{P}(LoS, \theta_{ku}))\kappa}{\gamma_0 ||\boldsymbol{b}_u - \boldsymbol{m}_k||^\alpha} \\ &= \frac{\hat{\mathbb{P}}(LoS, \theta_{ku})}{\gamma_0 ||\boldsymbol{b}_u - \boldsymbol{m}_k||^\alpha}, \end{aligned} \tag{3}$$

where $\kappa < 1$ is the attenuation effect of the NLoS channel, $\alpha \geq 2$ is the path loss exponent and $\gamma_0 = (\frac{4\pi f_c}{c})^2$ is the reference free space path loss at a distance of 1 m with f_c being the carrier frequency of the G2A channel and c being the speed of light. Then, the uplink transmission rate (bits/s) between the user k and the UAV is

$$R_{ku} = W_{ku} \log_2\left(1 + \frac{\hat{\mathbb{P}}(LoS, \theta_{ku}) P_k}{W_{ku} N_0 \gamma_0 ||\boldsymbol{b}_u - \boldsymbol{m}_k||^\alpha}\right), \tag{4}$$

where W_{ku} is the communication bandwidth, P_k is the transmission power of user k, N_0 is the noise power spectral density and α is the path loss exponent of the G2A channel.

Due to the rich scattering environment in the ground, the ground-to-ground (G2G) channel between the user device and the terrestrial BS can be modeled as a Rayleigh fading channel, which consists of large-scale attenuation and time-varying small-scale fading. We denote the small-scale fading coefficient as $h_{\text{G2G}}(t)$ and $\mathbb{E}[|h_{\text{G2G}}(t)|^2] = 1$. The instantaneous capacity of the G2G channel is given by

$$R_{kn}(t) = W_{kn} \log_2\left(1 + \frac{|h_{\text{G2G}}(t)|^2 P_k}{W_{kn} N_0 \gamma_0 ||\boldsymbol{b}_n - \boldsymbol{m}_k||^\beta}\right), \tag{5}$$

which changes dynamically in the fading channel. Given an outage probability tolerance ε, the expected transmission rate (bits/s) of the G2G Rayleigh fading channel can be calculated as [27]

$$R_{kn} = W_{kn} \log_2\left(1 + \frac{F^{-1}(\varepsilon) P_k}{W_{kn} N_0 \gamma_0 ||\boldsymbol{b}_n - \boldsymbol{m}_k||^\beta}\right), \tag{6}$$

where $F(\cdot)$ is the cumulative distribution function (CDF) of the Rayleigh fading coefficient, which can be obtained from empirical measurements.

B. Localization Model

We consider the OTDoA positioning method to localize the ground UEs. To reduce the computational complexity of UE, each UE estimates its 3D location from the reference signals sent by four ANs, including all the three ground BSs and a UAV. We assume that the accurate locations of terrestrial BSs and UAVs are known, since UAVs can use altimeters and satellite systems to locate themselves. Each UE measures the time-of-arrival (ToA) of localization signals from the aerial and terrestrial ANs. The TDoA observations are the time difference between the ToA of a reference node and the ToAs of the remaining ANs. Suppose that all ANs are accurately synchronized in time and BS 1 is the reference node, device k estimates its location $\boldsymbol{m}_k$ by solving the TDoA equations:

$$\tau_{n1} = (||\boldsymbol{b}_n - \boldsymbol{m}_k|| - ||\boldsymbol{b}_1 - \boldsymbol{m}_k||)/c, \quad n \in \mathcal{U} \cup \{2, 3\}, \tag{7}$$

where c is the propagation speed of the signals and τ_{n1} is the TDoA between the n-th anchor and the reference node BS 1. There exist many linear and non-linear estimation algorithms to obtain $\boldsymbol{m}_k$ from the TDoA equations [28]. In this paper, we are interested in the theoretical bound of localization accuracy regardless of the specific estimation algorithm.

Without loss of generality, we analyze the performance of the OTDoA method by considering the narrowband positioning reference signal (NPRS) in the narrowband IoT (NB-IoT) system. NPRS is an orthogonal frequency division multiplexing (OFDM) based downlink reference signal occupying 180 kHz bandwidth (one LTE resource block). The variance of the ToA measurement using N_{sub} OFDM subframes is expressed as [29]

$$\sigma^2_{\text{ToA}} = \frac{T_s^2}{N_{sub} \cdot 8\pi^2 \cdot \sum_{s \in \mathcal{S}} \sum_{i \in \mathcal{N}_s} p_i^2 i^2} \cdot \frac{1}{\text{SNR}} = \frac{\psi}{\text{SNR}}, \tag{8}$$

where T_s and SNR are the symbol duration and signal-to-noise ratio of the received signal, respectively. In one localization period, we have N_{sub} OFDM subframes and each subframe have a set $\mathcal{S} = \{1, 2, \ldots, S\}$ of symbols containing NPRS signal. In symbol s, the subset of subcarriers containing NPRS signal is denoted as $\mathcal{N}_s$ and $p_i \in [0, 1]$ is the relative power weight of each subcarrier i. According to (8), the variance of the ToA measurement from UAV u to UE k is given by

$$\sigma^2_{uk} = \psi \frac{W N_0}{g_{ku} P_u}, \tag{9}$$

where W is the NPRS bandwidth, N_0 is the noise power density, P_u is the transmission power of the UAV, and g_{ku} is the power gain of G2A channel.

In the G2G channel, advanced signal processing algorithms are needed to eliminate the impacts of multi-path and NLoS

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.

propagation on the ToA measurement. The random estimation error is commonly modelled as an additional Gaussian noise term with variance σ^2_{NLoS}. The variance of the ToA measurements for the G2G channel can be derived as

$$\sigma^2_{nk} = \psi \frac{W N_0 \gamma_0 ||\boldsymbol{b}_n - \boldsymbol{m}_k||^\beta}{P_n} + \sigma^2_{\text{NLoS}}, \quad n \in \mathcal{N}, \tag{10}$$

where P_n is the transmission power of the BS n. Then, the covariance matrix $\mathbf{R}_{\text{TDoA}}$ of the TDoA observations in unit s^2 is given by [30]

$$\begin{aligned}\mathbf{R}_{\text{TDoA}} &= \begin{bmatrix} \text{var}(\tau_{21}) & \text{cov}(\tau_{21},\tau_{31}) & \text{cov}(\tau_{21},\tau_{u1}) \\ \text{cov}(\tau_{31},\tau_{21}) & \text{var}(\tau_{31}) & \text{cov}(\tau_{31},\tau_{u1}) \\ \text{cov}(\tau_{u1},\tau_{21}) & \text{cov}(\tau_{u1},\tau_{31}) & \text{var}(\tau_{u1}) \end{bmatrix} \\ &= \begin{bmatrix} \sigma^2_{1k}+\sigma^2_{2k} & \sigma^2_{1k} & \sigma^2_{1k} \\ \sigma^2_{1k} & \sigma^2_{1k}+\sigma^2_{3k} & \sigma^2_{1k} \\ \sigma^2_{1k} & \sigma^2_{1k} & \sigma^2_{1k}+\sigma^2_{uk} \end{bmatrix}, \end{aligned} \tag{11}$$

where $\text{var}(x)$ is the variance of variable x and $\text{cov}(x,y)$ is the covariance of variables x and y.

C. Localization Performance Metrics

The Fisher information matrix (FIM) quantifies the amount of information about the unknown parameter that measurement vector carries. The FIM for OTDoA positioning measurements in (7) is defined as

$$\mathbf{F} = \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H}, \tag{12}$$

where $\mathbf{R} = c^2 \cdot \mathbf{R}_{\text{TDoA}}$ is the covariance matrix of OTDoA positioning in unit m^2, and $\mathbf{H}$ is the Jacobian matrix of the TDoA equations

$$\mathbf{H} = \begin{bmatrix} \frac{\partial \tau_{21}}{\partial x_k} & \frac{\partial \tau_{21}}{\partial y_k} & \frac{\partial \tau_{21}}{\partial h_k} \\ \frac{\partial \tau_{31}}{\partial x_k} & \frac{\partial \tau_{31}}{\partial y_k} & \frac{\partial \tau_{31}}{\partial h_k} \\ \frac{\partial \tau_{u1}}{\partial x_k} & \frac{\partial \tau_{u1}}{\partial y_k} & \frac{\partial \tau_{u1}}{\partial h_k} \end{bmatrix}, \tag{13}$$

and the elements in the first and second rows of $\mathbf{H}$ can be derived as

$$\frac{\partial \tau_{n1}}{\partial x_k} = \frac{x^b_n - x_k}{||\boldsymbol{b}_n - \boldsymbol{m}_k||} - \frac{x^b_1 - x_k}{||\boldsymbol{b}_1 - \boldsymbol{m}_k||}, \tag{14a}$$

$$\frac{\partial \tau_{n1}}{\partial y_k} = \frac{y^b_n - y_k}{||\boldsymbol{b}_n - \boldsymbol{m}_k||} - \frac{y^b_1 - y_k}{||\boldsymbol{b}_1 - \boldsymbol{m}_k||}, \tag{14b}$$

$$\frac{\partial \tau_{n1}}{\partial h_k} = \frac{h^b_n - h_k}{||\boldsymbol{b}_n - \boldsymbol{m}_k||} - \frac{h^b_1 - h_k}{||\boldsymbol{b}_1 - \boldsymbol{m}_k||}, \tag{14c}$$

with $n = 2, 3$. The elements in the third row of $\mathbf{H}$ can be obtained from (14) by replacing $\boldsymbol{b}_n$ with $\boldsymbol{b}_u$. There are several localization performance metrics defined based on the FIM. A common metric of localization accuracy is CRLB, which represents the theoretical lower bound of mean square estimation error. CRLB can be expressed as a function of UE location and UAV location $L(\boldsymbol{m}_k, \boldsymbol{b}_u) = \text{Tr}\left(\mathbf{F}^{-1}\right)$, where $\text{Tr}(\cdot)$ means the trace of matrix. However, it is difficult to derive an analytical expression of the feasible UAV location with CRLB. In this paper, we adopt the *D-optimality* criterion to analyze the localization performance, which is a well-known performance metric of localization accuracy [31]. With the *D-optimality* metric, we will be able to find intuitive and meaningful geometrical characterizations of UAV deployment in Section IV. Specifically, *D-optimality* seeks to maximize the following opt-D value

$$\text{opt-}D = \det(\mathbf{F}) = \frac{\det(\mathbf{H})^2}{\det(\mathbf{R})}. \tag{15}$$

The determinant of $\mathbf{F}$ is inversely proportional to the uncertainty area of an unbiased location estimation [32]. Intuitively, a *D-optimality* design minimizes the uncertainty ellipsoid, or equivalently, minimize the average location estimation error.

III. Problem Formulation

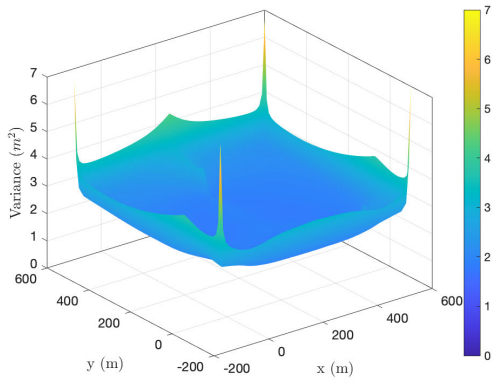
Before formulating the UAV deployment optimization problem, we first provide a motivating example of using UAV to assist the ground localization service. In particular, we show that the proposed UAV-assisted localization method can effectively improve the localization accuracy of the conventional scheme using only terrestrial BSs. For the ease of exposition, we divide an area $\mathcal{A} = [0, 600]$ m $\times$ $[0, 600]$ m into small boxes with width of 10 m each. Three BSs are deployed at location $(0, 0, 30)$, $(500, 0, 30)$ and $(250, 433, 30)$ with unit meter. At each candidate location, we optimize the horizontal placement of the fourth AN at $h = 30$ m and calculate the CRLB of the localization variance. For comparison, we place the UAV as the fourth AN at $h = 200$ m and optimize its horizontal location.[2]

After traversing all the points to be localized, we generate the heap maps in Fig. 2 according to the recorded estimation variance, when σ^2_{NLoS} is equal to 4×10^{-17} s^2. In Fig. 2, the x-y plane represents the horizontal coordinates of the localization points with height $h = 20$ m. The vertical axis is the minimum localization variance achievable by optimizing the location of the fourth AN, either the ground BS or UAV. Fig. 2(a) and Fig. 2(b) show the horizontal variance under both designs. Without the UAV, the horizontal variance is in the range of $[1.80, 7.12]$. With UAV as the additional AN, the horizontal variance reduced to the range of $[0.71, 2.45]$. The improvement is more significant in the vertical direction as shown in Fig. 2(c) and Fig. 2(d). The variance ranges are $[3.75, 5.63]$ and $[1.25, 1.88]$ for the conventional design and the UAV-assisted design, respectively.

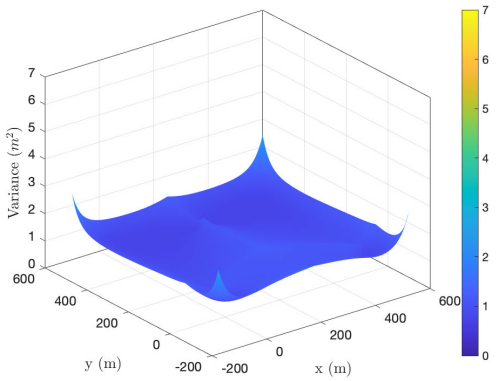
In Fig. 3, we observe the localization accuracy improvement and the conflicting impact of UAV placement. Fig. 3(a) shows the average variance under different NLoS noise. Each sample point is the average performance of 10 localization points randomly distributed in the area $\mathcal{A}$ with random height from the range $[10, 20]$ m. For both methods, the average variance will increase when the NLoS noise becomes larger. The UAV-assisted localization method produces much lower localization variance than the conventional scheme in both horizontal and vertical directions. The improvement is more significant in the vertical direction when σ^2_{NLoS} is large. For example, when σ^2_{NLoS} is equal to 10^{-15} s^2, the proposed

[2]Detailed simulation parameters are provided in the simulation section.

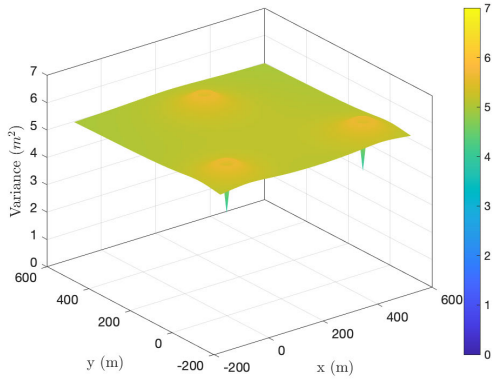
Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.



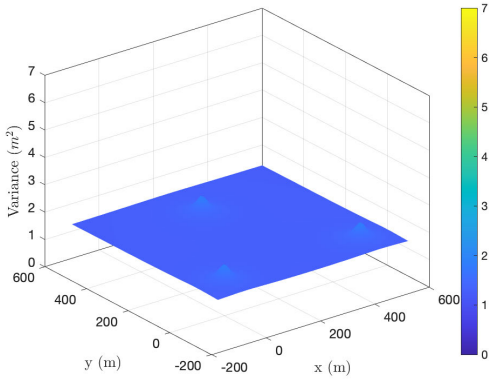
(a) Horizontal variance without UAV



(b) Horizontal variance with UAV

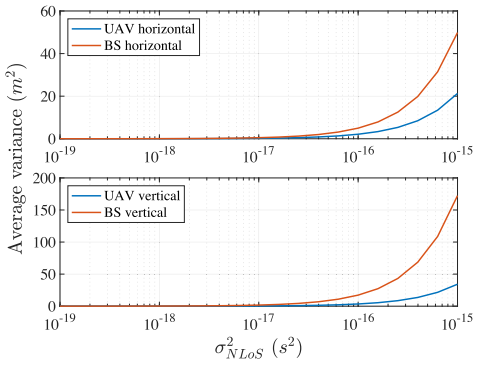


(c) Vertical variance without UAV

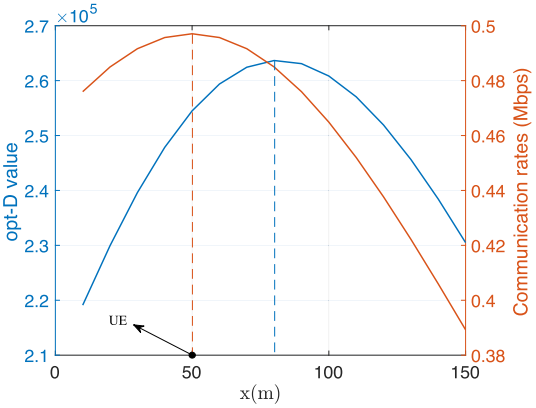


(d) Vertical variance with UAV

Fig. 2. Localization estimation variances under proposed design and conventional design, $\sigma^2_{NLoS} = 4 \times 10^{-17}$ s^2.



(a) Average localization variance versus σ^2_{NLoS}



(b) Localization and communication capabilities

Fig. 3. Variance and capability comparisons.

method reduces 57.3% horizontal variance and 80.1% vertical variance. To achieve localization accuracy at 1 m precision in both horizontal and vertical directions, the σ^2_{NLoS} is required to be lower than 4×10^{-18} s^2 for the conventional method and 2.5×10^{-17} s^2 for the UAV-assisted method. The above observations demonstrate the advantage of integrating UAVs for accurate 3D localization, especially in the vertical dimension. That is, the amplitude of UAV AN provides not only strong LoS link quality, but also the critical diversity in AN locations.

With fixed terrestrial BSs, we aim to deploy a minimum number of UAVs to satisfy the localization and communication

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.

requirements of all UEs. The problem is formulated as

$$\mathcal{P}1: \min_{\{\boldsymbol{b}_u\}_{u\in\mathcal{U}}} \quad |\mathcal{U}|, \tag{16a}$$

$$\text{s.t.} \quad \max_{\{\boldsymbol{b}_u\}_{u\in\mathcal{U}}} \ \text{opt-}D(\boldsymbol{m}_k, \boldsymbol{b}_u) \geq \epsilon_k, \quad \forall k \in \mathcal{K}, \tag{16b}$$

$$\max_{n\in\mathcal{U}\cup\mathcal{N}} R_{kn} \geq R_k^{th}, \forall k \in \mathcal{K}, \tag{16c}$$

$$\boldsymbol{b}_u \in \mathcal{A}, \forall u \in \mathcal{U}, \tag{16d}$$

where $|\mathcal{U}| = U$ is the number of UAVs and $\mathcal{A}$ is the restricted area to deploy the UAVs. Constraint (16b) is the localization accuracy requirement specified by an individual threshold ϵ_k. Constraint (16c) means that the communication rate of UE k should be larger than the threshold R_k^{th}. The maximization term in (16b) means that each UE estimates its location using all the three ground BSs and only one UAV. Meanwhile, the maximization term in (16c) means that each UE transmits information to only one base station, either a UAV or a ground BS.

The optimization problem is a cardinality minimization, which is NP-hard in general. An intuitive methodology is to search feasible solutions given fixed cardinality. However, this does not reduce the difficulty because the localization constraints are non-convex in regard to the UAV positions. In addition, $\mathcal{P}1$ contains two implicit association problems between UEs and BSs or UAVs, which are hard combinatorial optimization problems in multi-UAV and multi-BS scenarios. Another major difficulty arises from the potential conflicting impact of UAV placement to the communication and localization performance. As an illustrating example, in Fig. 3(b), we constraint the UAV flying at $h = 200$ m along a straight line passing over a UE at $(50, 0, 0)$. We see that from $x = 50$ to 80 m, the communication rate between the UAV and UE drops while the localization accuracy improves.

Overall, $\mathcal{P}1$ is difficult to solve, and it is even difficult to determine whether there exists a feasible solution satisfying all the constraints given the number of UAVs. In the next section, we analyze the localization performance constraints (16b) and derive closed-form expressions of feasible UAV hovering region.

IV. Geometric Characterization of Localization Constraints

In this section, we characterize the localization performance constraint (16b) in a geometric form. This enables the design of efficient graph-based algorithm to solve $\mathcal{P}1$ in the next section.

A. Approximation of Opt-D Value

To facilitate the analysis, we propose in this subsection an approximation of the opt-D value for a tagged UE k. For simplicity, we drop the index k and denote σ_{nk} as σ_n. We calculate the determinant of the covariance matrix

$$\begin{aligned}\det(\mathbf{R}) &= c^6\sigma_1^2\sigma_2^2\sigma_3^2 + c^6(\sigma_1^2\sigma_3^2 + \sigma_1^2\sigma_2^2 + \sigma_2^2\sigma_3^2)\sigma_u^2 \\ &= D_1 + D_2 \cdot \frac{\psi}{\text{SNR}_u},\end{aligned} \tag{17}$$

where $D_1 \triangleq c^6\sigma_1^2\sigma_2^2\sigma_3^2, D_2 \triangleq c^6(\sigma_1^2\sigma_3^2 + \sigma_1^2\sigma_2^2 + \sigma_2^2\sigma_3^2)$. Supposing that $D_1 \gg D_2 \cdot \frac{\psi}{\text{SNR}_u}$, we can ignore the second term and approximate $\det(\mathbf{R})$ using D_1. Then, we denote the corresponding value as opt-D_1

$$\text{opt-}D_1 = \frac{\det(\mathbf{H})^2}{D_1}. \tag{18}$$

In contrast, when $D_1 \ll D_2 \cdot \frac{\psi}{\text{SNR}_u}$, we ignore the constant term D_1 and approximate $\det(\mathbf{R})$ using the second term. We denote this approximation as opt-D_2, where

$$\text{opt-}D_2 = \frac{\det(\mathbf{H})^2\psi}{D_2\text{SNR}_u}. \tag{19}$$

In the following lemma, we show the condition when opt-D_1 is an accurate approximation.

Lemma 1: When $\frac{\text{SNR}_u}{\text{SNR}_n} \gg 3 - \frac{\sigma^2_{\text{NLoS}}}{\sigma_u^2}, \forall n \in \mathcal{N}$, opt-$D$ converges to opt-D_1.

Proof: The condition implies

$$\begin{aligned}&\frac{\text{SNR}_u}{\text{SNR}_n} + \frac{\sigma^2_{\text{NLoS}}\text{SNR}_u}{\psi} \gg 3 \\ &\Rightarrow \frac{\psi}{\text{SNR}_n} + \sigma^2_{\text{NLoS}} \gg \frac{3\psi}{\text{SNR}_u} \Rightarrow \sigma_n^2 \gg 3\sigma_u^2, \ \forall n \in N \\ &\Rightarrow \frac{\sigma_1^2\sigma_2^2\sigma_3^2}{3} \gg \sigma_2^2\sigma_3^2\sigma_u^2, \frac{\sigma_1^2\sigma_2^2\sigma_3^2}{3} \gg \sigma_1^2\sigma_3^2\sigma_u^2, \\ &\quad \frac{\sigma_1^2\sigma_2^2\sigma_3^2}{3} \gg \sigma_1^2\sigma_2^2\sigma_u^2.\end{aligned} \tag{20}$$

Summing up both sides of the three inequalities, we have

$$\sigma_1^2\sigma_2^2\sigma_3^2 \gg (\sigma_1^2\sigma_3^2 + \sigma_1^2\sigma_2^2 + \sigma_2^2\sigma_3^2)\sigma_u^2 \Leftrightarrow D_1 \gg D_2 \cdot \frac{\psi}{\text{SNR}_u}, \tag{21}$$

which is sufficient for opt-D converging to opt-D_1. ■

In emergency scenarios, the random estimation error σ^2_{NLoS} could be larger than the noise variance of G2A channel σ_u^2 due to the blockage and scattering effect in G2G channel. In Fig. 4(a), we compare the two approximations and the opt-D value with different ratios $\frac{\text{SNR}_u}{\text{SNR}_n} \in [10^{-2}, 10]$, and depict the values in a log scale. We observe that opt-D_1 is much more accurate than opt-D_2 within the considered ratio range. In Fig. 4(b), the opt-D and opt-D_1 values are almost on top with each other when the ratio is greater than 1, and the gap is less that 2% when the ratio is larger than 0.1. Therefore, we can safely adopt opt-D_1 as an accurate approximation in the following analysis.

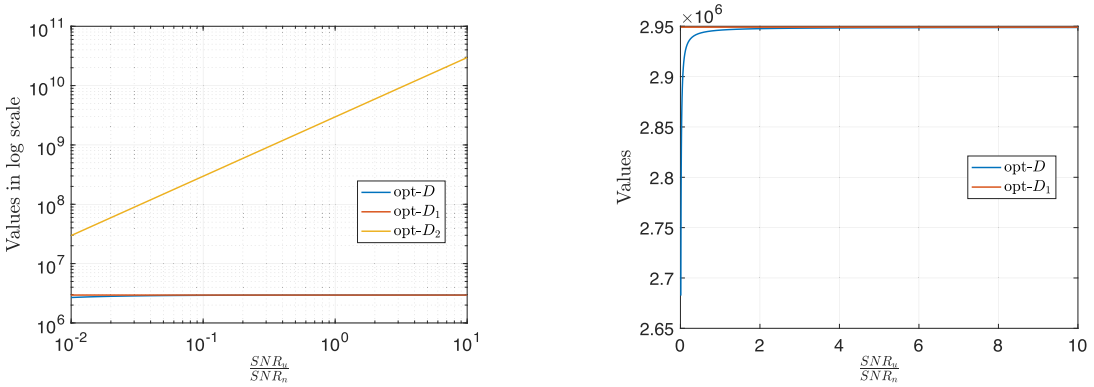
B. Equivalent Geometric Characterization

Firstly, we define the unit vector from the u-th UAV to a location point $\boldsymbol{m} = [x_0, y_0, h_0]$ as $\boldsymbol{q}_u$, and that from ground AN n to $\boldsymbol{m}$ as $\boldsymbol{q}_n$, where

$$\boldsymbol{q}_u = \frac{\boldsymbol{b_u} - \boldsymbol{m}}{||\boldsymbol{b_u} - \boldsymbol{m}||} = [q_{u1}, q_{u2}, q_{u3}], \quad \forall u \in \mathcal{U}, \tag{22}$$

$$\boldsymbol{q}_n = \frac{\boldsymbol{b_n} - \boldsymbol{m}}{||\boldsymbol{b_n} - \boldsymbol{m}||} = [q_{n1}, q_{n2}, q_{n3}], \quad \forall n \in \mathcal{N}. \tag{23}$$

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.



(a) The logarithm of $\det(\mathbf{R})$ versus the SNR ratio

(b) opt-D_1 and opt-D

Fig. 4. Approximations of opt-D values.

Then, $\mathbf{H}$ can be represented by the unit vectors

$$\mathbf{H} = \begin{bmatrix} \boldsymbol{q}_2 - \boldsymbol{q}_1 \\ \boldsymbol{q}_3 - \boldsymbol{q}_1 \\ \boldsymbol{q}_u - \boldsymbol{q}_1 \end{bmatrix}, \tag{24}$$

and we explicitly express $\det(\mathbf{H})$ using the elements in (14) such that

$$\det(\mathbf{H}) = \alpha_1 \left(\frac{x_u^a - x_0}{||\boldsymbol{b}_u - \boldsymbol{m}||} - q_{11} \right) + \alpha_2 \left(\frac{y_u^a - y_0}{||\boldsymbol{b}_u - \boldsymbol{m}||} - q_{12} \right) + \alpha_3 \left(\frac{h_u^a - h_0}{||\boldsymbol{b}_u - \boldsymbol{m}||} - q_{13} \right), \tag{25}$$

where the coefficients α_n for $n = 1, 2, 3$ are derived using (23)

$$\alpha_1 = (q_{22} - q_{12})(q_{33} - q_{13}) - (q_{23} - q_{13})(q_{32} - q_{12}), \tag{26}$$

$$\alpha_2 = (q_{23} - q_{13})(q_{31} - q_{11}) - (q_{21} - q_{11})(q_{33} - q_{13}), \tag{27}$$

$$\alpha_3 = (q_{21} - q_{11})(q_{32} - q_{12}) - (q_{22} - q_{12})(q_{31} - q_{11}). \tag{28}$$

The localization requirement opt-$D \geq \epsilon$ with opt-D_1 approximation in (18) becomes

$$\frac{\det(\mathbf{H})^2}{D_1} \geq \epsilon. \tag{29}$$

Based on the value of $\det(\mathbf{H})$, we derive the implicit geometry of (29) in two cases.

1) Case 1: When $\det(\mathbf{H}) \geq 0$, constraint (29) is equivalent to

$$\alpha_1(x_u^a - x_0) + \alpha_2(y_u^a - y_0) + \alpha_3(h_u^a - h_0) \geq \tilde{\epsilon}_1 ||\boldsymbol{b}_u - \boldsymbol{m}||, \tag{30}$$

where $\tilde{\epsilon}_1 = \sqrt{D_1 \epsilon} + \alpha_1 q_{11} + \alpha_2 q_{12} + \alpha_1 q_{13}$. The constraint (30) is a standard second-order cone constraint of dimension 4, and it is equivalent to

$$\alpha_1 q_{u1} + \alpha_2 q_{u2} + \alpha_3 q_{u3} \geq \tilde{\epsilon}_1. \tag{31}$$

Geometrically, the relative UAV location $\boldsymbol{q}_u = [q_{u1}, q_{u2}, q_{u3}]$ that satisfies (31) can be viewed as the intersection of the sphere $\mathbb{S} = \{[x, y, h] : x^2 + y^2 + h^2 = 1\}$ and the half space $\mathbb{H} = \{[x, y, h] : \alpha_1 x + \alpha_2 y + \alpha_3 h \geq \tilde{\epsilon}_1\}$. The normal vector of the plane $\mathbb{P} = \{[x, y, h] : \alpha_1 x + \alpha_2 y + \alpha_3 h = \tilde{\epsilon}_1\}$ is $\boldsymbol{n} = [\alpha_1, \alpha_2, \alpha_3]$ and the perpendicular distance from the origin to the plane is $\rho = \frac{|\tilde{\epsilon}_1|}{\sqrt{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}}$. The intersecting circle has radius $r = \sqrt{1 - \rho^2}$ and its center is

$$\boldsymbol{c}_0 = \rho \frac{\boldsymbol{n}}{||\boldsymbol{n}||} = \rho \frac{[\alpha_1, \alpha_2, \alpha_3]}{\sqrt{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}} = \frac{|\tilde{\epsilon}_1|}{\alpha_1^2 + \alpha_2^2 + \alpha_3^2} [\alpha_1, \alpha_2, \alpha_3]. \tag{32}$$

For the sphere and plane to have an intersection, it must hold that $\rho \leq 1$, i.e., $|\tilde{\epsilon}_1| \leq \sqrt{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}$. Accordingly, we establish in the following proposition a condition to check the feasibility of localization requirement.

Proposition 1: When $\det(\mathbf{H}) \geq 0$, the localization requirement for UE at $\boldsymbol{m}$ is feasible if and only if the value of ϵ satisfies

$$0 \leq \epsilon \leq \frac{(c_1 - c_2)^2}{D_1}, \tag{33}$$

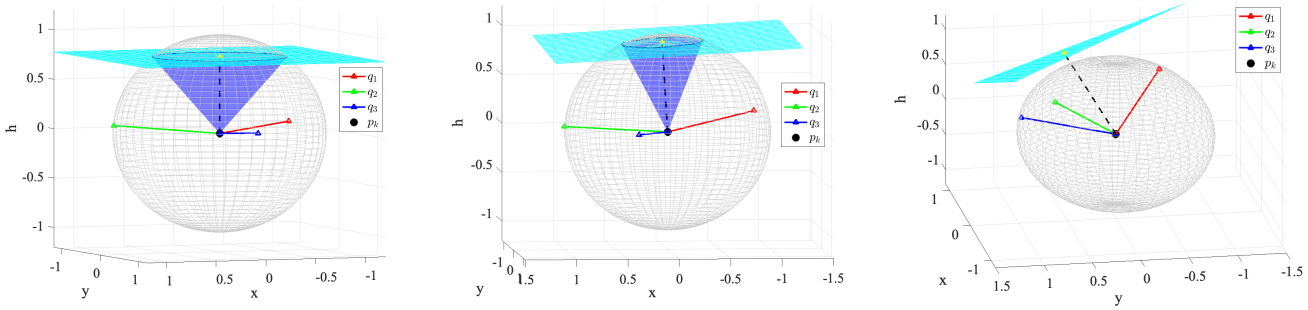
where c_1 and c_2 are given by

$$c_1 = \sqrt{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}, \quad c_2 = \alpha_1 q_{11} + \alpha_2 q_{12} + \alpha_3 q_{13}. \tag{34}$$

Proof: Please see the detailed proof in Appendix A. ■

Proposition 1 shows that by setting a reasonably small ϵ specified by (33), we can always find a UAV location to satisfy the localization accuracy requirement. Otherwise, if ϵ is beyond the range, it could be the case that no UAV placement can achieve the required accuracy level. Fig. 5 showcases that the feasibility of different UE location points under the same settings. For UE location $(250, 135, 10)$, the localization requirement is feasible when $\epsilon \leq 0.12$; for UE location $(100, 50, 10)$, it is feasible when $\epsilon \leq 0.081$; for UE location $(0, 10, 10)$, it is feasible when $\epsilon \leq 7.7 \times 10^{-3}$. As shown in Fig. 5, under the same $\epsilon = 8 \times 10^{-3}$, the sphere and the plane has intersection for UE locations $(250, 135, 10)$ and $(100, 50, 10)$, which means that the localization requirement is feasible for some UAV deployment solution. In Fig. 5(c), the sphere and the plane has no intersection, showing that no UAV deployment can satisfy the localization accuracy requirement for UE location $(0, 10, 10)$ when $\epsilon = 8 \times 10^{-3}$. Proposition 1 provides a convenient method to set a proper ϵ for the constraint (16b) to be feasible. Suppose that all the localization constraints (16b) are feasible by Proposition 1.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.



(a) $\boldsymbol{m} = (250, 135, 10)$ (b) $\boldsymbol{m} = (100, 50, 10)$ (c) $\boldsymbol{m} = (0, 10, 10)$

Fig. 5. Feasibility of different location points under $\epsilon = 8 \times 10^{-3}$.

We continue to derive in the following proposition the feasible UAV hovering region for (16b) assuming the UAV flies at a fixed altitude h_f.

Proposition 2: For a fixed altitude h_f, the localization constraint (29) is satisfied if and only if the UAV $\boldsymbol{b}_u = [x_u^a, y_u^a, h_f]$ lies within an ellipse defined by the following polynomial equation

$$\begin{aligned}\mathcal{E}^1: \ & (\frac{\alpha_1^2}{\tilde{\epsilon}_1^2} - 1)(x_u^a - x_0)^2 + \frac{2\alpha_1\alpha_2}{\tilde{\epsilon}_1^2}(x_u^a - x_0)(y_u^a - y_0) \\ & + (\frac{\alpha_2^2}{\tilde{\epsilon}_1^2} - 1)(y_u^a - y_0)^2 + \frac{2\alpha_1\alpha_3 h_r}{\tilde{\epsilon}_1^2}(x_u^a - x_0) \\ & + \frac{2\alpha_2\alpha_3 h_r}{\tilde{\epsilon}_1^2}(y_u^a - y_0) + (\frac{\alpha_3^2}{\tilde{\epsilon}_1^2} - 1)h_r^2 = 0, \end{aligned} \tag{35}$$

where $h_r = h_f - h_0$ is the relative altitude. The center $\boldsymbol{c}_1^e$ of the ellipse, the rotation angle θ_1, the semi-major axis l_1^a and the semi-minor axis l_1^i are respectively given by

$$\boldsymbol{c}_1^e = [\frac{\alpha_1\alpha_3 h_r}{\tilde{\epsilon}_1^2 - \alpha_1^2 - \alpha_2^2} + x_0, \frac{\alpha_2\alpha_3 h_r}{\tilde{\epsilon}_1^2 - \alpha_1^2 - \alpha_2^2} + y_0, h_f], \tag{36}$$

$$\theta_1 = \arctan(-\frac{\alpha_1}{\alpha_2}), \tag{37}$$

$$l_1^a = \sqrt{\frac{h_r^2(c_1^2 - \tilde{\epsilon}_1^2)}{\tilde{\epsilon}_1^2 - \alpha_1^2 - \alpha_2^2}}, \quad l_1^i = \frac{h_r\tilde{\epsilon}_1\sqrt{c_1^2 - \tilde{\epsilon}_1^2}}{\tilde{\epsilon}_1^2 - \alpha_1^2 - \alpha_2^2}. \tag{38}$$

Proof: We substitute $h_u^a = h_f$ to let the equality in (30) hold. Then, we square both sides of (30) and reformulate it into (35). For an ellipse with the general form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0, \tag{39}$$

its center $\boldsymbol{c}^e$, rotation angle θ^r, semi-major axis l^a and semi-minor axis l^i are given by

$$\boldsymbol{c}^e = [\frac{2CD - BE}{B^2 - 4AC}, \frac{2AE - BD}{B^2 - 4AC}], \tag{40}$$

$$\theta^r = \arctan\left(\frac{C - A - \sqrt{(A-C)^2 + B^2}}{B}\right), \tag{41}$$

$$l^a, l^i = \frac{\sqrt{2G(A + C \pm \sqrt{(A-C)^2 + B^2})}}{4AC - B^2}, \tag{42}$$

with $G = AE^2 + CD^2 - BDE + (B^2 - 4AC)F$. By substituting (35) into (39)-(42), we obtain (36)-(38). ■

Fig. 6(a) shows the 3D intersection of the extended cone and the plane at $h_f = 100$ m when $\boldsymbol{m} = (100, 50, 10)$. Fig. 6(b) shows the 2D projection of the feasible UAV hovering region. We also denote the center, the rotation angle, the semi-major axis and the semi-minor axis of the ellipse in Fig. 6(b). In Fig. 6(a), the UAV must be placed within the cone to satisfy the localization performance constraint. In Fig. 6(b), given a fixed altitude, the UAV must be placed within the eclipse to meet the localization requirement. Although we derive the feasible hovering region for any given h_f in Proposition 2, the distance between UAV and point $\boldsymbol{m}$ cannot be too large for two reasons. Firstly, when h_f is too large, the ellipse and its center defined in (35) and (36) could deviate from the UAV restricted area $\mathcal{A}$. In this case, we cannot find a feasible UAV location in $\mathcal{A}$. Secondly, the strength of localization signal from UAV to UE becomes very weak when the distance is too large, and thus the condition for an accurate opt-D_1 approximation no longer holds.

2) Case 2: When $\det(\mathbf{H}) < 0$, constraint (29) is equivalent to

$$\alpha_1(x_u^a - x_0) + \alpha_2(y_u^a - y_0) + \alpha_3(h_u^a - h_0) < \tilde{\epsilon}_2||\boldsymbol{b}_u - \boldsymbol{m}||, \tag{43}$$

where $\tilde{\epsilon}_2 = -\sqrt{D_1\epsilon} + \alpha_1 q_{11} + \alpha_2 q_{12} + \alpha_1 q_{13}$. Similar to the first case, constraint (43) is a standard second-order cone constraint.

Proposition 3: When $\det(\mathbf{H}) < 0$, the localization requirement for UE at $\boldsymbol{m}$ is feasible if and only if ϵ satisfies

$$0 \le \epsilon \le \frac{(c_1 + c_2)^2}{D_1}, \tag{44}$$

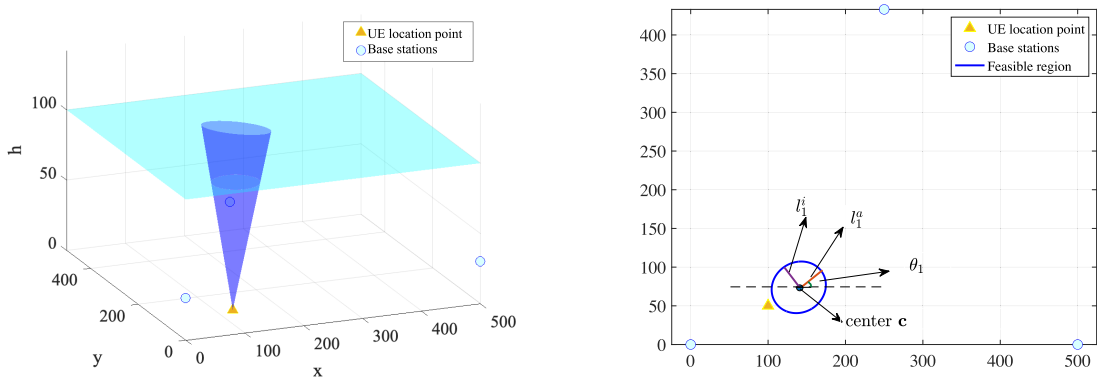
where c_1 and c_2 are the same as in (34).

Proof: The proof is similar to that of Proposition 1, and thus we omit it here. ■

Proposition 4: For a fixed altitude h_f, the localization constraint (43) is satisfied if and only if the UAV $\boldsymbol{b}_u = [x_u^a, y_u^a, h_f]$ lies within an ellipse defined by the following polynomial equation

$$\begin{aligned}\mathcal{E}^2: \ & (\frac{\alpha_1^2}{\tilde{\epsilon}_2^2} - 1)(x_u^a - x_0)^2 + \frac{2\alpha_1\alpha_2}{\tilde{\epsilon}_2^2}(x_u^a - x_0)(y_u^a - y_0) \\ & + (\frac{\alpha_2^2}{\tilde{\epsilon}_2^2} - 1)(y_u^a - y_0)^2 + \frac{2\alpha_1\alpha_3 h_r}{\tilde{\epsilon}_2^2}(x_u^a - x_0) \end{aligned}$$

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.



(a) 3D intersection of plane $h = h_f$ and cone (30)

(b) 2D projection of feasible region

Fig. 6. Intersections of $h_f = 100$ and the extended cone when $\boldsymbol{m} = (100, 50, 10)$.

$$+ \frac{2\alpha_2\alpha_3 h_r}{\tilde{\epsilon}_2^2}(y_u^a - y_0) + (\frac{\alpha_3^2}{\tilde{\epsilon}_2^2} - 1)h_r^2 = 0. \quad (45)$$

The center $\boldsymbol{c}_2^e$ of the ellipse, the rotation angle θ_2, the semi-major axis l_2^a and the semi-minor axis l_2^i are respectively given by

$$\boldsymbol{c}_2^e = [\frac{\alpha_1\alpha_3 h_r}{\tilde{\epsilon}_2^2 - \alpha_1^2 - \alpha_2^2} + x_0, \quad \frac{\alpha_2\alpha_3 h_r}{\tilde{\epsilon}_2^2 - \alpha_1^2 - \alpha_2^2} + y_0, h_f], \quad (46)$$

$$\theta_2 = \arctan(-\frac{\alpha_1}{\alpha_2}), \quad (47)$$

$$l_2^a = \sqrt{\frac{h_r^2(c_1^2 - \tilde{\epsilon}_2^2)}{\tilde{\epsilon}_2^2 - \alpha_1^2 - \alpha_2^2}}, \quad l_2^i = \frac{h_r\tilde{\epsilon}_2\sqrt{c_1^2 - \tilde{\epsilon}_2^2}}{\tilde{\epsilon}_2^2 - \alpha_1^2 - \alpha_2^2}. \quad (48)$$

Proof: By substituting (45) into (39)-(42), we obtain (46)-(48). ∎

In general, the overall feasible hovering region of a UAV can be specified by the union of two eclipses in Proposition 2 and 4, or just one of them. To simplify the algorithm design to solve $\mathcal{P}1$ in the next section, we prove in the following corollary that we can set a proper value ϵ for each UE such that the feasible hovering region of the UAV is specified by only one of the two eclipses in (35) and (45).

Corollary 1: By setting ϵ such that $|c_3 - |c_2|| \le \sqrt{D_1\epsilon} \le c_3 + |c_2|$ with $c_3 = \sqrt{\alpha_1^2 + \alpha_2^2}$, the feasible hovering region of the UAV is $\mathcal{E}^1$ in (35) if $c_2 \ge 0$. Otherwise, it is $\mathcal{E}^2$ in (45) if $c_2 < 0$.

Proof: Please see the detailed proof in Appendix B. ∎

V. Proposed Solution

By replacing the opt-D value with opt-D_1, we transfer $\mathcal{P}1$ into the following optimization problem

$$\mathcal{P}2: \min_{\{\boldsymbol{b}_u\}_{u\in\mathcal{U}}} \quad |\mathcal{U}|, \quad (49a)$$

$$\text{s.t.} \quad \max_{\{\boldsymbol{b}_u\}_{u\in\mathcal{U}}} \text{opt-}D_1(\boldsymbol{m}_k, \boldsymbol{b}_u) \ge \epsilon, \quad \forall k \in \mathcal{K}, \quad (49b)$$

$$\max_{n\in\mathcal{U}\cup\mathcal{N}} R_{kn} \ge R_k^{th}, \quad \forall k \in \mathcal{K}, \quad (49c)$$

$$\boldsymbol{b}_u \in \mathcal{A}, \quad \forall u \in \mathcal{U}. \quad (49d)$$

In this section, we propose a low-complexity algorithm to solve $\mathcal{P}2$. In addition to the feasible localization regions specified by Corollary 1, we also need to determine the feasible UAV region to meet communication requirements. According to the communication rate of G2A channel in (4), the communication requirement for UEs is equivalent to

$$||\boldsymbol{b}_u - \boldsymbol{m}_k||^\alpha \le \frac{P_k\hat{\mathbb{P}}(LoS, \theta_{ku})}{W_{ku}N_0\gamma_0(2^{R_k^{th}/W_{ku}} - 1)}. \quad (50)$$

Because $\hat{\mathbb{P}}(LoS, \theta_{ku})$ is also a function of $||\boldsymbol{b}_u - \boldsymbol{m}_k||$, it is difficult to depict (50) in graph. To efficiently characterize the feasible region, we restrict the communication requirement by finding a $\hat{\theta}_{ku}$ such that it is irrelevant to $||\boldsymbol{b}_u - \boldsymbol{m}_k||$ and satisfies $\hat{\mathbb{P}}(LoS, \hat{\theta}_{ku}) \le \hat{\mathbb{P}}(LoS, \theta_{ku})$.

Proposition 5: Let

$$\hat{\theta}_{ku} = \frac{180}{\pi}\arcsin\left(|h_f - h_k|\left(\frac{W_{ku}N_0\gamma_0(2^{R_k^{th}/W_{ku}} - 1)}{P_k}\right)^{\frac{1}{\alpha}}\right), \quad (51)$$

it holds that $\hat{\mathbb{P}}(LoS, \hat{\theta}_{ku}) \le \hat{\mathbb{P}}(LoS, \theta_{ku}), \forall\theta_{ku} \in [0°, 90°]$.

Proof: Please see the detailed proof in Appendix C. ∎

Then, the restricted communication requirement of UE k in G2A channels becomes

$$||\boldsymbol{b}_u - \boldsymbol{m}_k|| \le \left(\frac{P_k\hat{\mathbb{P}}(LoS, \hat{\theta}_{ku})}{W_{ku}N_0\gamma_0(2^{R_k^{th}/W_{ku}} - 1)}\right)^{\frac{1}{\alpha}}. \quad (52)$$

Notice that (50) is automatically satisfied if (52) holds. Recall that the communication rate of G2G channel is R_{kn} in (6). Then, BS n can meet the communication requirement of UE k if its location satisfies

$$||\boldsymbol{b}_n - \boldsymbol{m}_k|| \le \left(\frac{F^{-1}(\varepsilon)P_k}{W_{kn}N_0\gamma_0(2^{R_k^{th}/W_{ku}} - 1)}\right)^{\frac{1}{\beta}}. \quad (53)$$

Given a fixed altitude h_f, the feasible region of opt-$D_1(\boldsymbol{m}_k, \boldsymbol{b}_u) \ge \epsilon$ is an ellipse denoted by $\mathcal{E}_k$ and the feasible region of the communication requirement is a circle denoted by $\mathcal{C}_k$. We define a region set $\mathcal{R}$ as $\mathcal{R} = \bigcup_{k=1}^{K}(\mathcal{C}_k \cup \mathcal{E}_k)$, which includes all feasible regions of the communication and localization requirements. Firstly, we consider the communication requirement of each UE. If the location of UE k satisfies (53), BS n can already serve UE k's communication

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.

demand. We denote the UEs served by BSs as $\mathcal{K}_{BS}$ and their corresponding circles as $\mathcal{C}_{BS} = \bigcup_{k\in\mathcal{K}_{BS}} \mathcal{C}_k$. Then, when considering the placement of UAVs, we can exclude $\mathcal{C}_{BS}$ from $\mathcal{R}$. Therefore, the new set becomes $\mathcal{R}_{new} = \mathcal{R}\backslash\mathcal{C}_{BS}$. Our goal is to select a minimum-size subset $\mathcal{M} = \{\boldsymbol{b}_u\}_{u\in\mathcal{U}}$ of points such that every region in $\mathcal{R}_{new}$ has nonempty intersection with $\mathcal{M}$,. We recognize this as a classical minimum hitting set (MHS) problem, and summarize the solution algorithm to UAV deployment in Algorithm 1.

Algorithm 1 Solution Algorithm to UAV Deployment

Input: The UE locations $\{\boldsymbol{m}_k\}_{k=1}^K$; the fixed UAV altitude h_f;
Output: The required number of UAVs $|\mathcal{U}|$; the UAV locations $\mathcal{M} = \{\boldsymbol{b}_u\}_{u\in\mathcal{U}}$
Initialization: transmission power, bandwidth, R_k^{th}, $\mathcal{K}_{BS} = \varnothing$,$\mathcal{R} = \varnothing$;
Use Proposition 1 and 3 to set feasible values of ϵ_k;
for $k = 1$ to K **do**
  Obtain $\mathcal{C}_k$ according to (52) and determine $\mathcal{E}_k$ using (35) or (45);
  $\mathcal{R} = \mathcal{R} \cup \mathcal{C}_k \cup \mathcal{E}_k$;
  if $\boldsymbol{m}_k$ satisfies (53) **then**
    $\mathcal{K}_{BS} = \mathcal{K}_{BS} \cup \{k\}$; $\mathcal{C}_{BS} = \mathcal{C}_{BS} \cup \{\mathcal{C}_k\}$;
  end if
end for
Obtain $\mathcal{R}_{new} = \mathcal{R}\backslash\mathcal{C}_{BS}$;
Obtain $|\mathcal{U}|, \mathcal{M}$ by solving the MHS problem for $\mathcal{R}_{new}$;
Return: $|\mathcal{U}|, \mathcal{M}$

The MHS problem can be equivalently transferred into an integer liner programming (ILP) problem. The restricted area $\mathcal{A}$ is first discretized into a set of grid points $\mathcal{G} = \{\boldsymbol{g}_1, \boldsymbol{g}_2, \ldots, \boldsymbol{g}_L\}$, which are candidates for the deployment of UAVs. Given the candidate UAV points and the $\mathcal{R}_{new}$, we use a binary variable $v_{ij} = 1$ to denote that a grid point $\boldsymbol{g}_i$ is in set element $S_j \in \mathcal{R}_{new}$, where S_j could be a circle or an ellipse region. Otherwise, we set v_{ij} to be zero. Then, the ILP problem is formulated as

$$\mathcal{P}3: \min_{v_{ij}} \quad \sum_{i=1}^{L}\sum_{j=1}^{|\mathcal{R}_{new}|} v_{ij} \tag{54a}$$

$$\text{s.t.} \quad \sum_{i=1}^{L} v_{ij} \geq 1, \quad \forall S_j \in \mathcal{R}_{new}. \tag{54b}$$

We explain the ILP problem using a simple case with two UEs as shown in Fig. 7. The red circles are the feasible communication regions and the blue ellipses are the feasible localization regions. Therefore, the region set is $\mathcal{R}_{new} = \{\mathcal{C}_1, \mathcal{C}_2, \mathcal{E}_1, \mathcal{E}_2\}$. The small blue triangles are the candidate points $\mathcal{G} = \{\boldsymbol{g}_1, \boldsymbol{g}_2, \ldots, \boldsymbol{g}_6\}$ to deploy the UAVs.[3] The ILP problem becomes

$$\min_{v_{ij}} \quad \sum_{i=1}^{6}\sum_{j=1}^{4} v_{ij} \tag{55a}$$

$$\text{s.t.} \quad \sum_{i=1}^{6} v_{ij} \geq 1, \quad \forall S_j \in \{\mathcal{C}_1, \mathcal{C}_2, \mathcal{E}_1, \mathcal{E}_2\}, \tag{55b}$$

[3]The candidate points are randomly selected in space for illustration. In practice, the candidate points could be predetermined or calculated based on environment conditions and parameters.

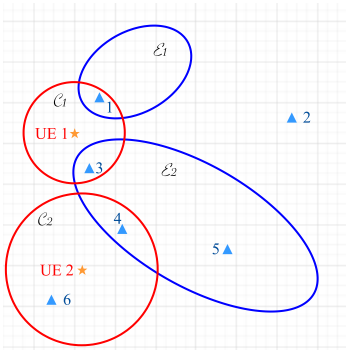


Fig. 7. Explanation with two UEs.

and the optimal solution is to deploy two UAVs at $\boldsymbol{g}_1$ and $\boldsymbol{g}_4$. By doing so, all the four circles/eclipses are hit by the two points.

The ILP problem is NP-hard and requires high computational complexity to obtain the optimal solution. Thus, we propose a Depth-first (DF) algorithm to solve the minimum hitting set problem. Firstly, we define the depth of point $\boldsymbol{g}_l$ as Δ_l, which is equal to the number of regions containing point $\boldsymbol{g}_l$, i.e., $|\mathcal{R}_l|$. Then, we select the grid point with the largest depth as the first UAV location. We remove the regions containing $\boldsymbol{g}_l$ and update the depth of grid points. The procedure is repeated until all regions are covered by at least one UAV. We use Fig. 7 to illustrate the algorithm. To start with, we observe that $\boldsymbol{g}_1$, $\boldsymbol{g}_3$ and $\boldsymbol{g}_4$ have the largest depth 2. Suppose that we firstly select $\boldsymbol{g}_1$ in $\mathcal{M}$. Since $\boldsymbol{g}_1$ is in $\mathcal{C}_1$ and $\mathcal{E}_1$, we remove these two regions from $\mathcal{R}_{new}$ and update the depth of grid points. Then, we see that $\boldsymbol{g}_4$ has the largest depth as it is inside the regions $\mathcal{C}_2$ and $\mathcal{E}_2$. Therefore, we find out that the solution is to deploy UAVs at $\boldsymbol{g}_1$ and $\boldsymbol{g}_4$. Nonetheless, the proposed solution may lead to suboptimal solution, e.g., selecting $\boldsymbol{g}_3$ in the first step would lead to deploy 3 UAVs at $\boldsymbol{g}_1$, $\boldsymbol{g}_3$ and $\boldsymbol{g}_4$. Still, we can improve the performance by executing the algorithm multiple times using randomized selection method and selecting the one produces the best solution.

The computation complexity of Algorithm 2 is $\mathcal{O}(LK + K^2)$, where L is the number of grid points and K is the number of UEs. The complexity can be further reduced by removing points not in any regions, i.e., $\Delta_l = 0$, and the number of candidate points will be significantly deducted. The DF algorithm is summarized in Algorithm 2.

VI. Simulation Results

In this section, we conduct simulation experiments to verify our analysis and evaluate the performance of our proposed algorithm. The simulation parameters are summarized in Table I unless otherwise stated. We compare our proposed algorithm with the following benchmark methods:

1) Integer Linear Programming (ILP) Method: The ILP problem $\mathcal{P}3$ is NP-hard and requires high computational complexity to obtain the optimal solution. We use the solution of ILP method as baseline to evaluate the performance of our proposed method.

2) Communication First Method: In this method, we firstly deploy the UAVs the satisfy the communication requirements

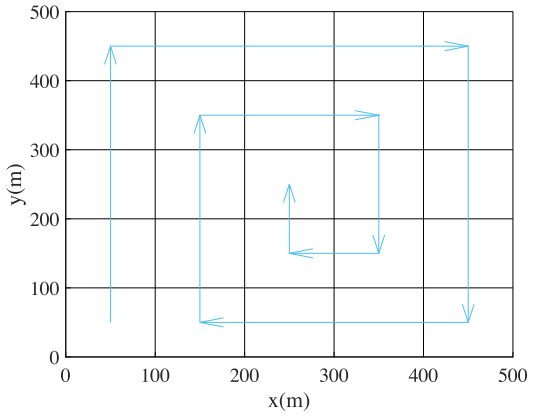
Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 Depth-First Algorithm

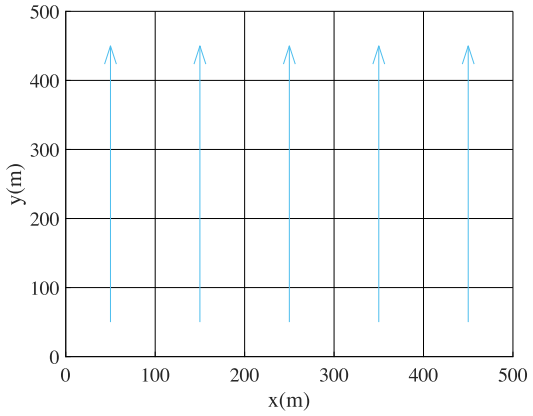
Input: $\mathcal{R}_{new}$, grid points $\mathcal{G} = \{\boldsymbol{g}_1, \boldsymbol{g}_2, \ldots, \boldsymbol{g}_L\}$;
Output: $|\mathcal{U}|, \mathcal{M}$;
Initialization: $\mathcal{M} = \varnothing$;
repeat
  for $l = 1$ to L **do**
    Find all regions in $\mathcal{R}_{new}$ containing point $\boldsymbol{g}_l$ and store them into $\mathcal{R}_l$;
    The depth of point $\boldsymbol{g}_l$ is $\Delta_l = |\mathcal{R}_l|$;
  end for
  Find one $\boldsymbol{g}_l$ with the largest depth Δ_l;
  $\mathcal{M} = \mathcal{M} \cup \{\boldsymbol{g}_l\}$;
  Remove the regions containing $\boldsymbol{g}_l$ from $\mathcal{R}_{new}$, $\mathcal{R}_{new} = \mathcal{R}_{new} \backslash \{S_i | S_i \ni \boldsymbol{g}_l, \forall S_i \in \mathcal{R}_{new}\}$;
until $\mathcal{R}_{new} = \varnothing$
$|\mathcal{U}| = |\mathcal{M}|$;
Return: $|\mathcal{U}|, \mathcal{M}$;

TABLE I
SIMULATION PARAMETERS

System Parameter	Value
Main frequency, f_c	2.1 GHz
Free space path loss at 1 m, γ_0	38.89 dB
Noise power spectral density, N_0	-174 dbm/Hz
Reference signal bandwidth, W	180 kHz
G2A path loss exponent, α	2
G2G path loss exponent, β	2.2
Transmission power of the BSs, P_n	0.1 W
Transmission power of the UAV, P_u	0.1 W
Transmission power of the UEs, P_k	0.01 W
UE uplink transmission bandwidth, W_{ku}, W_{kn}	180 kHz
Maximum tolerable outage probability ε	0.1
Inverse of the CDF, $F^{-1}(\varepsilon)$	0.11
Environmental parameter, a	15
Environmental parameter, b	0.5
Symbol duration of the OFDM signal, T_s	$\frac{1}{1.5\times10^4}$ s
Number of subframes containing NPRS	160
Location of BS 1, $\boldsymbol{b}_1$	$[100, 100, 30]$
Location of BS 2, $\boldsymbol{b}_2$	$[250, 433, 30]$
Location of BS 3, $\boldsymbol{b}_3$	$[500, 250, 30]$



(a) Spiral searching method



(b) Strip searching method

Fig. 8. Heuristic searching methods.

of all UEs. The deployed UAVs can satisfy the localization requirements of some UEs. Afterwards, we deploy additional UAVs to meet the localization demands of the remaining UEs.

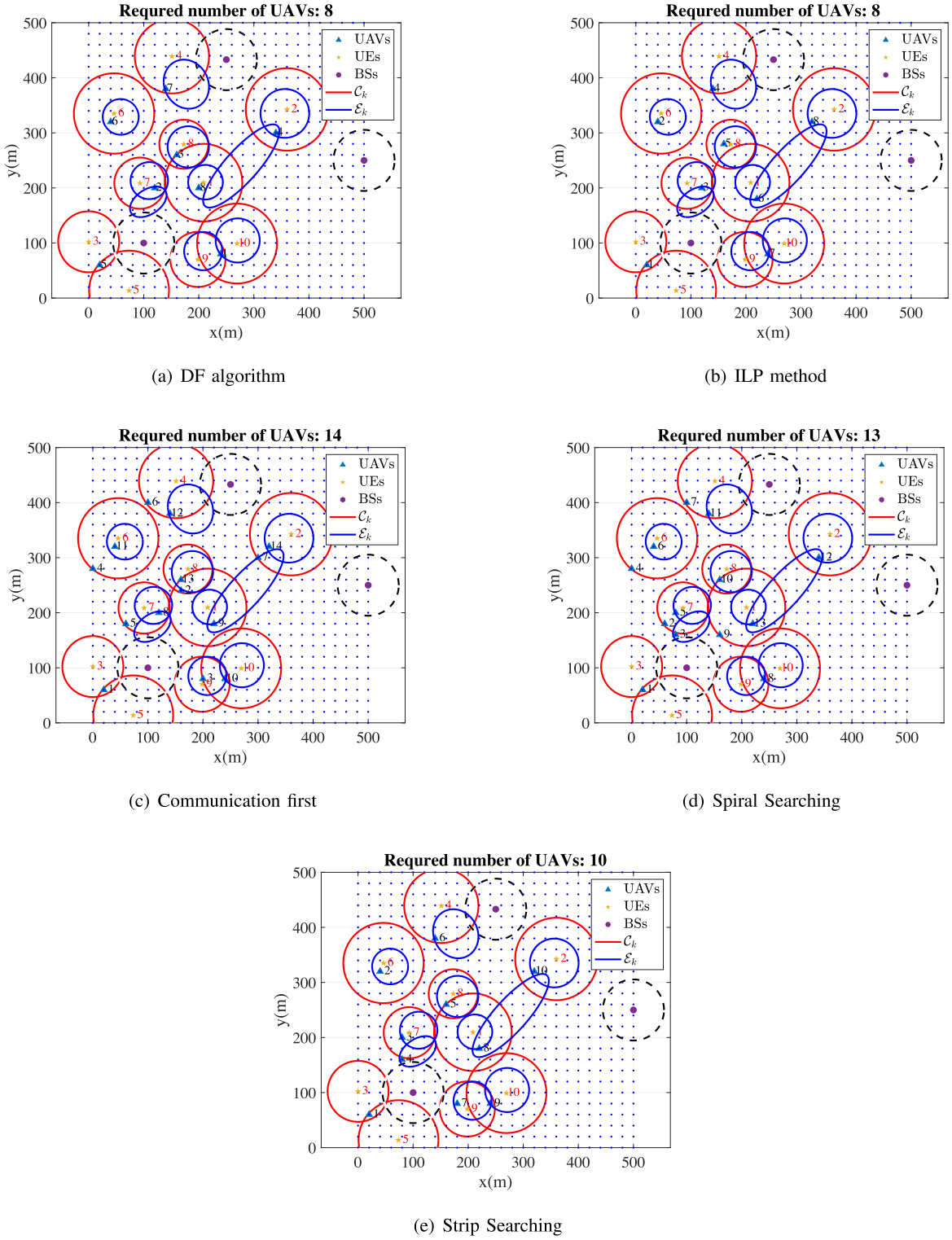
3) Spiral Searching Method: In [16], the authors proposed a polynomial-time algorithm to sequentially deploy the UAVs along a spiral path toward the center to provide wireless communication coverage to ground users. Similarly, we deploy the UAVs to satisfy the communication and localization requirements of all UEs along a simplified spiral path shown in Fig. 8(a).

4) Strip Searching Method: In [33], the authors proposed a near-linear-time approximation algorithm for the hitting set problem under specific geometric settings. The main idea is to use vertical decomposition to obtain overlapping regions. We adopt the strip searching as a benchmark method, which deploys the UAVs in the sequence shown in Fig. 8(b) from left to right.

We discretize the area $\mathcal{A}$ with equal spacing $\Delta = 20\ m$. Consequently, we obtain $L = 26 \times 26 = 676$ grid points for the deployment of UAVs. Fig. 9 shows the required numbers and the placements of UAVs under different methods for 10 UEs. The localization accuracy requirement ϵ_k is randomly selected from the range specified by Corollary 1 to ensure feasibility. The communication requirement set as $R_k^{th} = 2.77$ Mbps. Each UE is associated with one circular region (red circle) to meet the communication performance and an elliptical region (blue ellipse) to meet the localization performance. The dashed black circles are the communication coverage of the three BSs. As long as there is a UAV hovering in its corresponding blue area, its localization accuracy requirement is satisfied. Likewise, as long as there is a UAV or ground base station in the red area, the communication requirement can be satisfied. As shown in Fig. 9, the ILP method and the proposed DF method achieve the minimal number of UAVs, which is 8 in this scenario. The spiral searching method and the strip searching method require 13 UAVs and 10 UAVs, respectively. The communication first method requires 14 UAVs, which is the largest among all methods. Although we use the approximation opt-D_1 values to determine the feasible localization regions, the obtained UAV locations always satisfy the original localization requirement (16b) in opt-D values. Fig. 10(a) and Fig. 10(b) show that all methods can satisfy the communication and localization requirements of all UEs.

In Fig. 11, we show the average required number of UAVs under all methods from 2 UEs to 40 UEs. Each sample point is the average of 100 simulation runs. The ILP method attains the optimal solutions and provides the baseline for comparison.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.



(a) DF algorithm

(b) ILP method

(c) Communication first

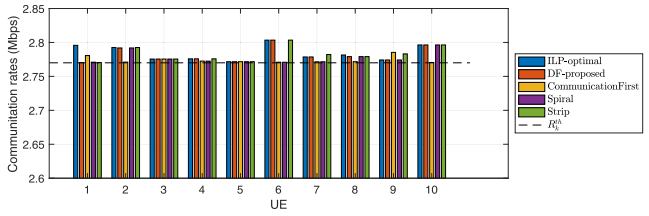
(d) Spiral Searching

(e) Strip Searching

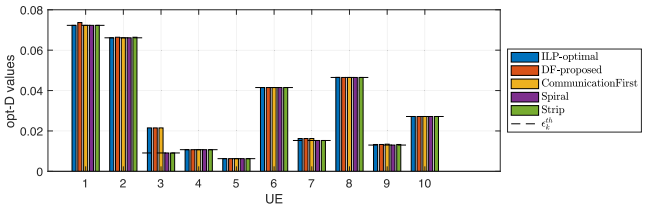
Fig. 9. Deployment of UAVs under different methods.

As shown in Fig. 11, the DF method is the closest to the ILP method. The communication first method and the spiral searching method require more than twice the optimal UAV numbers. The reason is that these methods are based on the communication coverage method. Therefore, they are not suitable for the UAV deployment for joint communication and localization scenarios. The performance of the strip searching method is slightly better than these two methods, which requires less than twice the optimal UAV numbers. When the number of UEs is 40, the average UAV number of the ILP method is 16.7. The DF method on average needs 1.95 more UAVs than the ILP method, while the strip searching method, the spiral searching method and the communication first method require 12.56, 17.77 and 18.98 more UAVs, respectively.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.



(a) Communications rates of 10 UEs



(b) opt-D values of 10 UEs

Fig. 10. Comparison of communication and localization performance under different UAV deployment methods.

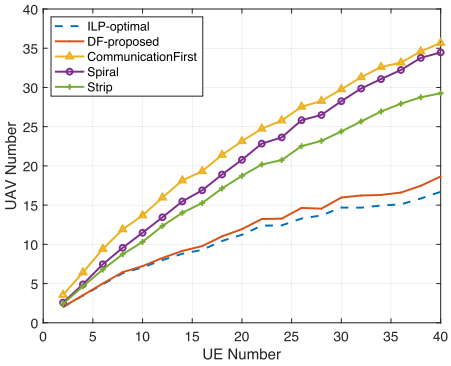


Fig. 11. UAV number v.s. UE number under different methods.

VII. Conclusion and Discussions

In this paper, we investigate the optimization problem of deploying the minimal number of UAVs in feasible regions to satisfy the communication and localization requirements of the ground users. By adopting D-optimality as the localization performance metric, we derive the closed-form expression and the spatial characteristics of the feasible UAV hovering regions. We solve the deployment problem by transforming the problem into a geometric minimum hitting set problem, and propose a low-complexity algorithm to solve it. We compare our proposed algorithm with benchmark methods. The number of UAVs required by the proposed algorithm is close to the optimal number, while the other benchmark methods require much more UAVs.

In the future, we consider to extend our analysis into the scenarios where multiple moving UAVs provide ILAC services to ground UEs. We will investigate the geometric structure of using multiple UAVs to assist localization. We will incorporate energy consumption constraints into the system design and propose energy-efficient algorithms to optimize the UAV trajectories. We will also analyze the impact of power allocation to improve the communication and localization performance.

Appendix A
Proof of Proposition 1

With $c_1 = \sqrt{\alpha_1^2+\alpha_2^2+\alpha_3^2}$ and $c_2 = \alpha_1 q_{11}+\alpha_2 q_{12}+\alpha_3 q_{13}$, we have $|c_2| \le c_1$ according to Cauchy-Schwartz inequality:

$$\begin{aligned}&|\alpha_1 q_{11}+\alpha_2 q_{12}+\alpha_3 q_{13}|\\ &\quad\le \sqrt{\alpha_1^2+\alpha_2^2+\alpha_3^2}\sqrt{q_{11}^2+q_{12}^2+q_{13}^2}\\ &\quad\le \sqrt{\alpha_1^2+\alpha_2^2+\alpha_3^2}||\boldsymbol{q}_1|| = \sqrt{\alpha_1^2+\alpha_2^2+\alpha_3^2}.\end{aligned} \tag{56}$$

The feasible condition for the intersection of sphere and plane is $\rho \le 1$. Therefore, we have

$$\begin{aligned}&\rho = \frac{|\tilde{\epsilon}_1|}{c_1} \le 1 \Rightarrow |\tilde{\epsilon}_1| \le c_1 \Rightarrow |\sqrt{D_1\epsilon}+c_2| \le c_1\\ &\Rightarrow -c_1-c_2 \le \sqrt{D_1\epsilon} \le c_1-c_2.\end{aligned} \tag{57}$$

Since $|c_2| \le c_1$, we have $-c_1-c_2 \le 0$ and (57) becomes $\sqrt{D_1\epsilon} \le c_1-c_2$.

Appendix B
Proof of Corollary 1

The general conic equation $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$ represents an ellipse if and only if

$$B^2-4AC<0. \tag{58}$$

For (35), condition (58) is equivalent to

$$\frac{4\alpha_1^2\alpha_2^2}{\tilde{\epsilon}_1^2}-4\left(\frac{\alpha_1^2}{\tilde{\epsilon}_1^2}-1\right)\left(\frac{\alpha_2^2}{\tilde{\epsilon}_1^2}-1\right)<0 \Leftrightarrow \frac{\alpha_1^2+\alpha_2^2}{\tilde{\epsilon}_1^2}-1<0. \tag{59}$$

Since $\tilde{\epsilon}_1 = \sqrt{D_1\epsilon}+c_2$ and we define $c_3 = \sqrt{\alpha_1^2+\alpha_2^2} \ge 0$, the condition is equivalent to

$$\begin{aligned}&|\sqrt{D_1\epsilon}+c_2| > c_3\\ &\quad\Leftrightarrow \sqrt{D_1\epsilon} > c_3-c_2, \text{ or } \sqrt{D_1\epsilon} < -c_3-c_2,\end{aligned} \tag{60}$$

Similarly, because $\tilde{\epsilon}_2 = -\sqrt{D_1\epsilon}+c_2$, the condition (58) for (45) is equivalent to

$$\begin{aligned}&|-\sqrt{D_1\epsilon}+c_2| > c_3,\\ &\quad\Leftrightarrow \sqrt{D_1\epsilon} > c_2+c_3, \text{ or } \sqrt{D_1\epsilon} < c_2-c_3.\end{aligned} \tag{61}$$

When $c_2 \ge 0$, $\sqrt{D_1\epsilon} < -c_3-c_2 \le 0$ in (60) can be discarded. If ϵ satisfies $|c_3-c_2| < \sqrt{D_1\epsilon} \le c_3+c_2$, (60) holds while (61) does not hold. When $c_2 < 0$, $\sqrt{D_1\epsilon} < c_2-c_3 < 0$ in (61) can be ignored. If $|c_3+c_2| < \sqrt{D_1\epsilon} \le c_3-c_2$, (61) holds while (60) does not hold.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.

APPENDIX C
PROOF OF PROPOSITION 5

Since $\hat{\mathbb{P}}(LoS, \theta_{ku}) \leq 1$, we have

$$\begin{aligned} ||\boldsymbol{b}_u - \boldsymbol{m}_k||^{\alpha} &\leq \frac{P_k \hat{\mathbb{P}}(LoS, \theta_{ku})}{W_{ku} N_0 \gamma_0 (2^{R_k^{th}/W_{ku}} - 1)} \\ &\leq \frac{P_k}{W_{ku} N_0 \gamma_0 (2^{R_k^{th}/W_{ku}} - 1)} \\ \Rightarrow ||\boldsymbol{b}_u - \boldsymbol{m}_k|| &\leq \Big(\frac{P_k}{W_{ku} N_0 \gamma_0 (2^{R_k^{th}/W_{ku}} - 1)}\Big)^{\frac{1}{\alpha}} = R_0 \\ \Rightarrow \frac{|h_f - h_k|}{||\boldsymbol{b}_u - \boldsymbol{m}_k||} &\geq \frac{|h_f - h_k|}{R_0} \Rightarrow \sin\theta_{ku} \geq \sin\hat{\theta}_{ku}. \end{aligned} \tag{62}$$

In addition, both $\sin\theta_{ku}$ and $\hat{\mathbb{P}}(LoS, \theta_{ku})$ are monotonically increasing when $\theta_{ku} \in [0°, 90°]$, thus $\sin\theta_{ku} \geq \sin\hat{\theta}_{ku}$ implies $\hat{\mathbb{P}}(LoS, \theta_{ku}) \geq \hat{\mathbb{P}}(LoS, \hat{\theta}_{ku})$. Hence, the proof is complete.

REFERENCES

[1] C. Hu et al., "Digital twin-assisted real-time traffic data prediction method for 5G-enabled Internet of Vehicles," *IEEE Trans. Ind. Informat.*, vol. 18, no. 4, pp. 2811–2819, Apr. 2022.

[2] L. Liu, X. Guo, and C. Lee, "Promoting smart cities into the 5G era with multi-field Internet of Things (IoT) applications powered with advanced mechanical energy harvesters," *Nano Energy*, vol. 88, Oct. 2021, Art. no. 106304.

[3] M. Kumhar and J. Bhatia, "Emerging communication technologies for 5G-enabled Internet of Things applications," in *Blockchain for 5G-Enabled IoT*. New York, NY, USA: Springer, 2021, pp. 133–158.

[4] J. Qiu, D. Grace, G. Ding, J. Yao, and Q. Wu, "Blockchain-based secure spectrum trading for unmanned-aerial-vehicle-assisted cellular networks: An operator's perspective," *IEEE Internet Things J.*, vol. 7, no. 1, pp. 451–466, Jan. 2020.

[5] M. M. Hasan et al., "Search and rescue operation in flooded areas: A survey on emerging sensor networking-enabled IoT-oriented technologies and applications," *Cogn. Syst. Res.*, vol. 67, pp. 104–123, Jun. 2021.

[6] R. Shahzadi, M. Ali, H. Z. Khan, and M. Naeem, "UAV assisted 5G and beyond wireless networks: A survey," *J. Netw. Comput. Appl.*, vol. 189, Sep. 2021, Art. no. 103114.

[7] X. Zhang, X. Tao, F. Zhu, X. Shi, and F. Wang, "Quality assessment of GNSS observations from an Android N smartphone and positioning performance analysis using time-differenced filtering approach," *GPS Solutions*, vol. 22, no. 3, pp. 1–11, Jul. 2018.

[8] Z. Wang et al., "Toward reliable UAV-enabled positioning in mountainous environments: System design and preliminary results," *IEEE Trans. Rel.*, vol. 71, no. 4, pp. 1435–1463, Dec. 2022.

[9] Z. Xiao and Y. Zeng, "An overview on integrated localization and communication towards 6G," *Sci. China Inf. Sci.*, vol. 65, no. 3, pp. 1–46, Mar. 2022.

[10] A. Xhafa, J. A. del Peral-Rosado, J. A. López-Salcedo, and G. Seco-Granados, "Evaluation of 5G positioning performance based on UTDoA, AoA and base-station selective exclusion," *Sensors*, vol. 22, no. 1, p. 101, Dec. 2021.

[11] S. Fischer, "Observed time difference of arrival (OTDOA) positioning in 3GPP LTE," *Qualcomm White Paper*, vol. 1, no. 1, pp. 1–62, 2014.

[12] C.-Y. Chen and W.-R. Wu, "Three-dimensional positioning for LTE systems," *IEEE Trans. Veh. Technol.*, vol. 66, no. 4, pp. 3220–3234, Apr. 2017.

[13] J. Qiu, D. Grace, G. Ding, M. D. Zakaria, and Q. Wu, "Air-ground heterogeneous networks for 5G and beyond via integrating high and low altitude platforms," *IEEE Wireless Commun.*, vol. 26, no. 6, pp. 140–148, Dec. 2019.

[14] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal LAP altitude for maximum coverage," *IEEE Wireless Commun. Lett.*, vol. 3, no. 6, pp. 569–572, Dec. 2014.

[15] Y. Chen, W. Feng, and G. Zheng, "Optimum placement of UAV as relays," *IEEE Commun. Lett.*, vol. 22, no. 2, pp. 248–251, Feb. 2018.

[16] J. Lyu, Y. Zeng, R. Zhang, and T. J. Lim, "Placement optimization of UAV-mounted mobile base stations," *IEEE Commun. Lett.*, vol. 21, no. 3, pp. 604–607, Mar. 2017.

[17] M. A. Ali and A. Jamalipour, "UAV placement and power allocation in uplink and downlink operations of cellular network," *IEEE Trans. Commun.*, vol. 68, no. 7, pp. 4383–4393, Jul. 2020.

[18] H. Ren, C. Pan, K. Wang, W. Xu, M. Elkashlan, and A. Nallanathan, "Joint transmit power and placement optimization for URLLC-enabled UAV relay systems," *IEEE Trans. Veh. Technol.*, vol. 69, no. 7, pp. 8003–8007, Jul. 2020.

[19] R. Fan, J. Cui, S. Jin, K. Yang, and J. An, "Optimal node placement and resource allocation for UAV relaying network," *IEEE Commun. Lett.*, vol. 22, no. 4, pp. 808–811, Apr. 2018.

[20] J. Sabzehali et al., "3D placement and orientation of mmWave-based UAVs for guaranteed LoS coverage," *IEEE Wireless Commun. Lett.*, vol. 10, no. 8, pp. 1662–1666, Aug. 2021.

[21] J. A. D. Peral-Rosado, R. Raulefs, J. A. López-Salcedo, and G. Seco-Granados, "Survey of cellular mobile radio localization methods: From 1G to 5G," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 2, pp. 1124–1148, 2nd Quart., 2017.

[22] Y. Liu, J. Wang, and Y. Shen, "Spectrum allocation for UAV-aided relative localization of ground vehicles," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, May 2018, pp. 1–5.

[23] Z. Wu, W. Quan, and T. Zhang, "Resource allocation in UAV-aided vehicle localization frameworks," in *Proc. IEEE/CIC Int. Conf. Commun. Workshops China (ICCC Workshops)*, Aug. 2019, pp. 98–103.

[24] F. B. Sorbelli, S. K. Das, C. M. Pinotti, and S. Silvestri, "Range based algorithms for precise localization of terrestrial objects using a drone," *Pervas. Mobile Comput.*, vol. 48, pp. 20–42, Aug. 2018.

[25] J. Yang, T. Liang, and T. Zhang, "Deployment optimization in UAV aided vehicle localization," in *Proc. IEEE 93rd Veh. Technol. Conf. (VTC-Spring)*, Apr. 2021, pp. 1–6.

[26] O. Esrafilian, R. Gangula, and D. Gesbert, "Three-dimensional-map-based trajectory design in UAV-aided wireless localization systems," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 9894–9904, Jun. 2021.

[27] C. Zhan, Y. Zeng, and R. Zhang, "Energy-efficient data collection in UAV enabled wireless sensor network," *IEEE Wireless Commun. Lett.*, vol. 7, no. 3, pp. 328–331, Jun. 2018.

[28] A. W. Al-Asadi and N. S. Ali, "Noise-robust least-squares method in TDOA estimation of a source location," in *Proc. Palestinian Int. Conf. Inf. Commun. Technol. (PICICT)*, Sep. 2021, pp. 123–128.

[29] J. A. D. Peral-Rosado, J. A. López-Salcedo, G. Seco-Granados, F. Zanier, and M. Crisci, "Achievable localization accuracy of the positioning reference signal of 3GPP LTE," in *Proc. Int. Conf. Localization GNSS*, Jun. 2012, pp. 1–6.

[30] Z. Wang, R. Liu, Q. Liu, J. S. Thompson, and M. Kadoch, "Energy-efficient data collection and device positioning in UAV-assisted IoT," *IEEE Internet Things J.*, vol. 7, no. 2, pp. 1122–1139, Feb. 2020.

[31] D. Ucinski, *Optimal Measurement Methods for Distributed Parameter System Identification*. Boca Raton, FL, USA: CRC Press, 2004.

[32] Y. Zheng, M. Sheng, J. Liu, and J. Li, "OpArray: Exploiting array orientation for accurate indoor localization," *IEEE Trans. Commun.*, vol. 67, no. 1, pp. 847–858, Jan. 2019.

[33] P. K. Agarwal, E. Ezra, and M. Sharir, "Near-linear approximation algorithms for geometric hitting sets," *Algorithmica*, vol. 63, nos. 1–2, pp. 1–25, Jun. 2012.



Zheyuan Yang (Graduate Student Member, IEEE) received the B.Eng. degree in information engineering from The Chinese University of Hong Kong in 2019, where she is currently pursuing the Ph.D. degree. Her research interests include optimizations in UAV-enabled mobile edge computing, UAV trajectory control, integrated sensing and communication, and waveform optimization for communication with sensing.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.



Suzhi Bi (Senior Member, IEEE) received the Ph.D. degree in information engineering from The Chinese University of Hong Kong in 2013. From 2013 to 2015, he was a Post-Doctoral Research Fellow with the Department of Electrical and Computer Engineering, National University of Singapore. Since 2015, he has been with the College of Electronics and Information Engineering, Shenzhen University, China, where he is currently an Associate Professor. His research interests include optimization and machine learning techniques for wireless information and power transfer, mobile computing, and wireless sensing. He received the 2019 IEEE ComSoc Asia-Pacific Outstanding Young Researcher Award, the 2021 IEEE ComSoc Asia-Pacific Outstanding Paper Award, and the Conference Best Paper Award of IEEE SmartGridComm 2013, IEEE/CIC ICCC 2021, and IEEE VTC-Spring 2022. He is an Editor of IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS and IEEE WIRELESS COMMUNICATIONS LETTERS.



Ying-Jun Angela Zhang (Fellow, IEEE) received the Ph.D. degree from the Department of Electrical and Electronic Engineering, The Hong Kong University of Science and Technology.

She joined the Department of Information Engineering, The Chinese University of Hong Kong, in 2005, where she is currently a Professor. Her research interests include optimization and learning in wireless communication systems.

Prof. Zhang is a Member-at-Large of the IEEE ComSoc Board of Governors and a member of the steering committees of IEEE TRANSACTIONS ON MOBILE COMPUTING, IEEE WIRELESS COMMUNICATION LETTERS, and IEEE SmartgridComm Conference. She was a co-recipient of the 2021 and 2014 IEEE ComSoc Asia Pacific Outstanding Paper Awards, the 2013 IEEE SmartgridComm Best Paper Award, and the 2011 IEEE Marconi Prize Paper Award on Wireless Communications. As the only winner from engineering science, she received the Hong Kong Young Scientist Award in 2006, conferred by the Hong Kong Institute of Science. She was the Founding Chair of the IEEE ComSoc Technical Committee of Smart Grid Communications. She served as the Editor-in-Chief for IEEE OPEN JOURNAL OF THE COMMUNICATIONS SOCIETY, the Chair of the Executive Editor Committee of IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, and many years on the editorial boards of IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE TRANSACTIONS ON COMMUNICATIONS, IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS Special Issues, IEEE INTERNET OF THINGS JOURNAL Special Issues, and *IEEE Communications Magazine* Special Issues. She has served on the organizing committees of many top conferences, such as IEEE GLOBECOM, ICC, VTC, and SmartgridComm.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:35:07 UTC from IEEE Xplore. Restrictions apply.